Turning Neighbourhood Word-of-Mouth into Portable Reputation

Research report on designing, building, securing, monetising and launching a locally trusted reputation and service-matching platform for informal artisans in Cameroon. Synthesised from nine research notes current to 12 November 2026. Not legal advice: items flagged "lawyer" need Cameroonian counsel.

Executive summary

Launch with **domestic electricians in Douala's northern residential belt (Bonamoussadi, Makepe, Kotto)**. Run it as a **WhatsApp-, USSD- and agent-first trust layer** on a **Python/FastAPI serverless stack on AWS**. Verification should be community-anchored: two vouches from different social circles plus a practical test judged by peer masters. Reputation should be computed **only from jobs the platform can prove happened**: an OTP or QR handshake at the job, a mobile-money payment reference, and in-app before/after photos. Customers should see conservative "verified jobs + % satisfied" figures rather than inflated star averages. Cameroon does not lack artisans. It lacks a trusted system: **86.6% of jobs are informal**, only **2.1% of informal workers have written contracts**, and customers find tradespeople through neighbours, phone numbers painted on poles, and luck (INS EESI3; Ecofin; MIT Solve: Maibeta).

At least seven Cameroonian startups have tried to digitise artisan hiring, and none has visibly reached scale. That points to liquidity and depth of verification as the bottleneck, not access to an app. Three legal constraints shape the build:

- **The platform must not hold customer funds.** Regulation 04/18/CEMAC/UMAC/COBAC reserves fund-holding to licensed institutions, so "escrow" has to sit in a licensed PSP's or aggregator's ring-fenced account, with the platform sending only release instructions.
- **Trust scores are a criminal-law risk until a lawyer clears them.** Data Protection Law 2024/017 criminalises "profiling" (Art. 65: 3–10 years). Scores must be consented to, based on service quality, explainable and open to human contest, and need a written legal opinion before automated scoring goes live.
- **Hosting is a cross-border transfer.** Cameroon has no AWS Region, so every transfer abroad legally needs prior authorisation from a data authority that, as of November 2026, has not been confirmed as operational.

The monetisation model that meets the "no upfront charge to artisans" rule has four parts:

- **A customer-paid protection fee** of about 8–10%.
- **A success commission** deducted only from payouts the artisan has already earned: 0% at launch, then 5–8% after the third paid job.
- **Higher-margin B2B contracts and diaspora milestone projects.** Household jobs contribute only about **XAF 650 (about €1) per job** at steady state, so these streams have to carry the business.
- **Ancillary revenue** from materials, credit and insurance referrals.

The report closes with a 12-week concierge pilot, expansion gates, a four-minute end-to-end demo, a register of open questions, and an 801-entry de-duplicated bibliography.

Key decisions at a glance

Decision	Recommendation	Main reason
Trade at launch	Domestic electricians (repairs, surge damage, solar/inverter wiring as add-on); plumbing in phase 2	Highest safety cost of a bad hire, frequent and urgent demand, quality the customer cannot judge
Place at launch	Bonamoussadi + Makepe + Kotto (Douala V)	Dense middle-class landlords and households, high artisan supply, no anglophone-region shutdown risk
Channels	Field agents + WhatsApp + USSD + SMS + printed QR/numeric badge first; IVR voice second; PWA third	53.7% of Cameroonians are offline; text menus fail first-time low-literacy users
Proof of skill	5-tier ladder: registered → known locally (2-circle vouches) → work shown (OTP-confirmed jobs) → skill tested (2 masters) → trusted pro; CQP/VAE as parallel badge	Copies the RPL pattern used across Africa at low cost
Rating	Binary thumbs + tags, Beta posterior with 12-month decay, conservative lower bound displayed	Resists inflation; removed a racial earnings gap in a 2025 *Nature* study
Payments	Licensed PSP/aggregator collections + payouts; PSP-held hold-and-release; cash still allowed but weighted lower	CEMAC 04/18 licensing
Backend	FastAPI on Lambda (Web Adapter) + Aurora PostgreSQL/PostGIS + H3 + DynamoDB + EventBridge/SQS/Step Functions	Near-zero idle cost; same container moves to Fargate at scale
Region	eu-west-3 (Paris) primary, eu-west-1 DR, cell-ready for af-south-1; decided by an RTT test plus legal opinion	Cable topology favours Europe; both options are legal transfers needing authorisation
Revenue	Customer fee + post-earning commission + B2B + diaspora + ancillary	Fits the constraint; household margins are thin

1. Cameroon's problem is a trust vacuum, not a skills shortage

A huge, atomised and invisible workforce

Informal work is the norm in Cameroon. The INS EESI3 survey (2021 fieldwork) puts **86.6% of all jobs in the informal sector**. Informal non-agricultural work is **67.5% of urban employment** (INS EESI3 page). There are more than **3.4 million informal production units**, 63.7% of them urban and 56.1% headed by women (Cameroon Tribune citing INS). The units are tiny: 78.3% are one-person operations. Average informal income is **XAF 83,409 a month**. Only 2.1% of workers have a written contract, and 50.2% rely on verbal ones (Ecofin).

Nobody knows how many artisans there are, and the official figures conflict:

- The crafts minister claims crafts are "more than 50%" of workers, with registered artisan production units rising from 11,814 (2019) to 72,508 (2025) (EcoMatin).
- The 2022 ministry yearbook counted only **5,912 registrations** that year (Journal du Cameroun).

These are probably different measures: annual flow versus cumulative stock. Neither comes close to the real population. That gap is the opening: no state register can tell a customer who is competent.

Supply is plentiful and under-used. Douala has the country's **highest unemployment (15.4%)**, national underemployment is **61.4%**, and only 1.1% of jobseekers use employment agencies (INS EESI3 full report). The binding constraint is matching and trust, not the number of workers.

How hiring works today, and how it fails

Customers find artisans through personal networks, the trainer (*patron*) an artisan learned under, numbers painted on poles, and walk-up workshops. Maibeta's founders summed up the market: households face "inconsistent pricing, limited accountability, and uncertainty about work quality". The core problem is "a lack of a trusted system", not a lack of skilled workers (TEF: Maibeta).

The local trust signals already exist offline. Buea apprentices say "when people know your trainer, they become confident in you… that's why you will always see certificates and pictures in most shops" (Frontiers in Education 2022). Classifieds such as CoinAfrique offer no verification beyond warnings not to send money before checking (CoinAfrique). That warning implies advance-payment fraud is common.

The documented failures fall into three groups. **Safety failures** are the most serious:

- Fires blamed on chaotic wiring, low-end equipment and power-cut surges are recurring in Yaoundé, Douala and Bafoussam, and Douala's Mboppi market burned at least three times in 2025 (Le360).
- In February 2026 a mother and five children died in a Yaoundé house fire, with a short circuit among the hypotheses (AllAfrica).
- No national register of domestic electrocutions exists, and experts urge people to hire "a certified electrician" that customers have no way to identify (Camer.be 2026).
- Building collapses killed 33 people in 2023 and 6 in **Bonamoussadi in July 2026**, the proposed pilot area (Journal du Cameroun; Camer.be).

Reliability failures include missed deadlines (Buea tailors admit they "do not meet deadlines") and abandoned sites. **Price failures** take the form of opaque, inconsistent quotes. One widely repeated claim, that "85% of fires are electrical", **could not be verified** on the source page and should not be used.

The state framework is shallow

Law No. 2007/004 governs crafts. Artisans register free at Communal Crafts Offices (BCA) for a card that requires an ID, a workshop plan and photos. The card verifies identity and location, not competence, and mainly unlocks fairs and training (Cameroon Desks). A dedicated Chamber of Trades has been planned without success. A 2026 op-ed argues registration is burdensome "without any immediate perceived benefit" (Camer.be op-ed).

Cameroonian competitors confirm demand but not traction:

- **QwickServices** launched in Douala in April 2026 with 2,000+ categories, escrow-style payment and the Fon of Mankon on its board (Cameroon Tribune).
- **Quicky** in Yaoundé reported about 500 downloads (WeAreTech).
- **OmniJobber** (2018) recruited providers door to door and charged 8–20% commission (MIT Solve).

Implication: compete on depth of verification and local vouching in one trade and one area, not on breadth of categories.

Why electricians in Douala-North

The trade choice is well supported by the evidence. Electrical work maximises the gap between how valuable a good reputation is and how hard quality is to observe:

- Wiring is hidden in walls, which makes it a classic "credence good".
- Failures can be fatal.
- Outages and surges make demand frequent and urgent.
- Urban electricity access is 87.6% (INS ECAM5).

The neighbourhood choice is a reasoned judgement, at moderate confidence, that a short field scan should validate. Bonamoussadi is Douala's main northern housing area and Kotto its middle-class hub (Wikipedia: Douala). Buea and Bamenda are ruled out for a first pilot by the anglophone conflict and the 2017 shutdowns: **93–94 days** offline, followed by about four more months of social-media restriction (Access Now).

Conflict resolved. The launch notes (09) use plumbing as their worked example, while the context note (01) and business note (08) favour electrical work. This report picks **electricians first and plumbing as the first adjacent trade**, because many Douala artisans advertise as "électricien-plombier" anyway, and electrical work offers both the stronger safety story and an emerging certification route.

2. Artisans prove skill through vouches, verified jobs and peer-judged tests

Copy the African RPL pattern, cheaply

Across Kenya, Ghana, Nigeria, South Africa and Benin, recognition of prior learning (RPL) follows the same structure: **practical demonstration, portfolio, screening interview, and an optional knowledge check**, with **trade associations as mobilisers and master artisans as assessors**. Three examples:

- Kenya's NITA runs awareness → screening → portfolio assessment → practical assessment → certification (NITA). Its jua kali federation identified the candidates (NITA, 8 counties).
- Benin's CQM **replaced the "diplôme de libération" previously issued by master artisans** (Gouv.bj). This shows that master-signed graduation already carries social legitimacy in francophone Africa.
- The ILO concludes that "bottom-up skills assessments are feasible and yield multiple benefits" if well managed (ILO 2015).

Cameroon has the legal basis but not yet the scale:

- A 2019 law added VAE (validation of prior learning) (La Base Lextenso), and Decree 2020/2592 tasks training establishments with promoting it (Décret 2020/2592).
- In July 2026 the ministries validated occupational standards and **candidate portfolios** for a construction-sector RPL framework. It is **not yet operational** (ILO, Aug 2026).
- The only active household-adjacent VAE route found is for mechanics, through CRAMA–MINEFOP (fundsforNGOs).

State certification can therefore only be a top badge held by a minority. It cannot be a gate to joining.

Staked, diverse, expiring vouches

Microfinance research shows social ties can substitute for collateral. Network connections work as collateral, and strong ties improve job recommendations (Karlan et al., QJE 2009). Giné and Karlan found, however, that repayment held up when joint liability was dropped, as long as public meetings continued (Giné & Karlan). Visible, repeated accountability does the work; financial co-liability is not needed.

Web3 identity systems add three engineering rules:

- vouches must come from diverse clusters (BrightID);
- credentials carry weights and expire (Human Passport stamps expire after 90 days) (Human Passport);
- transitive trust should be seeded from pre-trusted local anchors (EigenTrust) (EigenTrust).

In Cameroon, njangi groups already enforce trust through social exclusion (Fair Observer). Vigilance committees are a warning, though. Douala's post-2017 self-defence committees abused residents and had to be purged (Camer.be). Chiefs and committees should therefore count as **one signal, never a gate**, and every decision should be open to appeal.

Recommended verification ladder

Tier	Badge	Requirement	Signals
0 Registered	grey circle	Phone OTP; name matches mobile-money account; live selfie; trade and quarter declared	Real, reachable person
1 Known locally	green house	≥2 vouches from *different* circles: trade (master, association head) **and** residence/faith (chef de quartier, pastor/imam, women's-group leader); workshop checked once by agent	"People here know him/her"
2 Work shown	camera	≥5 in-app, server-timestamped, hash-unique job photos **and** ≥3 OTP-confirmed jobs from distinct customer numbers	Has recently done this work
3 Skill tested	hammer + tick	Recorded 30–90-minute standard task scored by **two** association masters on a 10–15-item checklist aligned to MINEFOP standards, **or** a verified CAP/CQP/DQP/VAE certificate	Peer-judged competence
4 Trusted pro	gold star	≥20 verified jobs over ≥6 months; conservative satisfaction above the local threshold; no unresolved disputes	Sustained reliability
Parallel	flag	Official MINEFOP qualification verified	Formal recognition and bridge to VAE

The ladder runs under four rules:

- **Vouch budget.** Each voucher holds at most five active vouches. An upheld fraud, abuse or no-show-with-deposit complaint within 12 months costs the voucher reputation. Two such events cost vouching rights. Vouches expire after 12 months.
- **No single gatekeeper.** Associations nominate assessors and earn a fee per test, but cannot grant Tier 1 or Tier 3 alone (the two-circle and two-master rules).
- **Decay.** Badges lapse without activity: Tier 2 needs at least one verified job per 90 days.
- **Inclusion.** Women and displaced anglophone artisans, who lost their home networks, can use a women's-group or faith leader as their residence-circle voucher and can ask for a female assessor.

The pilot should measure inter-rater agreement between the two masters (target kappa ≥0.6, an assumption) and whether the Tier 3 score predicts later complaints. Every numeric parameter here is a design starting point, not a value calibrated for Cameroon.

3. Customers trust what the platform can prove, shown simply

Only verified jobs count

Raw star averages fail everywhere:

- The median eBay seller shows **100% positive**, but "effective percent positive" averages only 64%, because unhappy buyers leave silently (Nosko & Tadelis, NBER).
- Airbnb ratings average about 4.7 (Zervas et al.).
- Ratings inflate over time as raters lower their standards (Filippas, Horton & Golden).

Booking-gated platforms show less manipulation than open ones (Mayzlin et al., NBER). The founding rule is therefore **no review without a verified job**.

Photo metadata is not proof on its own. EXIF timestamps can be edited and mock-location apps are legal (TrueScreen). Hardware-signed C2PA provenance exists on premium phones like the Pixel 10 but will not reach low-end Android soon (Google Security Blog). Proof must rest on **customer-side confirmation plus server-side checks**:

1. A customer-initiated booking.
2. A one-time code from the customer's phone (SMS, USSD or app), entered by the artisan.
3. Live-camera before/after photos only, hashed and server-timestamped with coarse location.
4. Where possible, a mobile-money reference tied to the job.

Urban Company adds an arrival selfie checked against the professional's identity, so that "jobs are not passed onto someone else" (Urban Company).

Binary ratings, a conservative score and simultaneous reveal

A 2025 *Nature* field study of 100,759 home-services jobs is decisive here. Under five-star ratings, non-white workers earned about 91 cents per dollar earned by white workers. **Switching to thumbs up/down eliminated the gap** (Rice University). Airbnb's simultaneous-reveal experiment raised review rates and produced more candid reviews (Fradkin et al.). Uber sets minimum ratings **per city** because rating cultures differ (Uber). And hiring newcomers produces information that benefits everyone, so the platform should subsidise first jobs (Pallais, NBER).

The recommended score works as follows:

- **Eligible feedback.** Only feedback from OTP- or payment-confirmed jobs counts, with at most one counted rating per customer phone per artisan per 30 days.
- **Format.** Thumbs up or down plus tap tags: on time, price kept, clean, respectful, would hire again.
- **Decay.** Each job carries a weight of $\exp(-\ln 2 \cdot \text{age}/365)$, a **12-month half-life**.
- **Silence counts.** Confirmed jobs with no feedback add $\lambda \approx 0.25$ to the negative count, as in the eBay "effective percent positive" logic.
- **Prior.** Beta($\alpha_0=4$, $\beta_0=1$).

- **Display.** The 10th percentile of Beta(α_0+R, β_0+S), shown as "% satisfied" beside the **count of verified jobs**. Jøsang's Beta reputation system and the Wilson lower bound both keep thin records from looking stronger than they are (Jøsang & Ismail; Evan Miller).
- **Reveal.** Both sides' ratings are revealed together after both submit or after 7 days.

Conflicts resolved. The notes disagree on three parameters:

- *Rating scale.* The channels note (05) proposes 1–5 smiley faces mapped to keypad digits, and the demo script (09) shows stars. The reputation note (03) argues for binary. This report keeps a **binary scored signal**, collected on USSD and IVR as "1 = 👍, 2 = 👎". Smiley faces stay as a visual aid only.
- *Review window.* The anti-fraud note (04) suggests 14–30 days; note 03 suggests 7. Use a **7-day reveal window** for ratings, since jobs are short, and a separate **30-day workmanship-guarantee window** for complaints.
- *Half-life.* Note 04's 12–18 months versus note 03's 12. Start at 12 and tune.

What the customer sees

For low-literacy users, text-free interfaces with graphics and voice are strongly preferred (Medhi et al.). Trust also transfers from known, linked parties (Stewart 2003).

The artisan card should show:

- a large face photo and a trade pictogram;
- 3–6 recent in-app job photos;
- "**34 jobs checked ✓**" and "**92% 👍**" (conservative);
- the top two tags as icons and the tier badge as a coloured shape;
- a moderated 15–30-second voice testimonial;
- a neighbour line ("**5 people from your quarter hired him**"), computed from verified jobs;
- a "Garantie" shield on first jobs.

Two warnings apply. First, never tie the verification badge to paid promotion. Jiji shows "Verified" only to sellers who buy boost packages, which empties the badge of meaning (Jiji FAQ). Second, voice testimonials need moderation and a no-reply option, because women on voice forums face harassment (Vashistha et al.).

In Cameroon specifically, a PLS-SEM survey of 273 households found that **trust built from repeated good experiences and operators visibly monitoring risk** drove continued mobile-money use more than institutional assurances did (Abertay University). The platform should therefore *show* that it is watching: a one-tap report button, fraud warnings, and visible action on reports.

Trust runs both ways

Fairwork audits in Kenya and Nigeria repeatedly found ratings "often biased towards customers" and deactivations without due process (Fairwork Kenya 2023; Fairwork Nigeria 2023). The platform should commit to five protections from day one:

- Artisans rate customers ("paid as agreed", "respectful", "safe site").
- Repeat non-payers must pay a deposit before booking.

- No automatic deactivation from ratings alone.
- Every decision comes with written reasons, a 14-day appeal and human review.
- A monthly bias audit (see section 10 for the data-protection limits on how it is done).

4. Fast, fair matching comes from a small ranked shortlist and published price bands

Retrieve, rank, explore, dispatch

The matching engine should be a transparent two-stage pipeline:

- **Retrieval.** A SQL query of under 50 ms pulls artisans with the right skill inside H3 resolution-8 rings k≤2–3 around the request (each cell is about 0.74 km² (H3 table)). Candidates must be active, verified to the tier the job category requires, not overloaded, and available.
- **Scoring.** score = w1·trust_lcb + w2·f(distance) + w3·responsiveness + w4·price_fit + w5·availability − w6·load + exploration_bonus. The weights live in config so ops can tune them per city.
- **Logging.** Every impression, contact, quote, hire and outcome is logged so the platform can move to learning-to-rank once it has thousands of outcomes.

Thumbtack moved from heuristics to ML and measured position bias using randomised allocation (Thumbtack, arXiv). Airbnb lifted new-listing bookings **14%** by replacing a default value for new listings with a learned estimate (Airbnb, arXiv). Fairness of exposure treats ranking as an obligation to the people being ranked (Singh & Joachims). Hence **one in five slots goes to an exploration pool**: new artisans who are verified and vouched but have no reviews, chosen by Thompson sampling. Cumulative exposure per artisan is capped so a few stars do not capture all demand.

Dispatch depends on urgency:

- **Scheduled jobs:** a shortlist of 3–5 the customer chooses from.
- **Urgent jobs** (sparking socket, total outage): offers go to the top 3 with a 2–5 minute acceptance window, then cascade to the next 3. Lynk gave pros 15 minutes to reply Yes/No by SMS and matched in 2–6 hours (GSMA: Lynk).
- **High concurrency:** above about 20 urgent requests per minute in a zone, switch to batched Hungarian assignment (SciPy `linear_sum_assignment`), the DoorDash "predictions plus optimisation" pattern (DoorDash).

Stable-matching algorithms are unnecessary. Neither side holds full preference lists.

Fair price without surge

Price opacity is a named pain point. Lynk found open quoting produced slow, inconsistent quotes, while standardised fixed-price packages gave customers a pro within two hours (The Flip: Lynk).

The pricing approach has three parts:

- **Price observations.** Record quoted and final prices per job category and H3 resolution-7 zone.
- **Published bands.** Show P25–P75 of final prices over 90 days, with a minimum of 20 jobs and fallback to city level ("typical: 8,000–15,000 FCFA"). Nudge quotes above P90 rather than blocking

them.

- **Standard packages.** Publish fixed price ranges in French and Pidgin for the ten most common jobs: install socket, replace breaker, earth a circuit, and so on.

The working price hypotheses (unverified, to validate in 30–50 interviews) are XAF 5,000–15,000 for an electrician call-out and small repair, and XAF 100,000–500,000+ for a full rewiring. **No dynamic surge at MVP.** Surge would damage trust in an informal market. Supply and demand per hexagon should drive *incentives* (extra exposure for artisans in under-supplied cells) instead (Uber H3).

Two service-level targets anchor the "fast" promise: median time-to-match **≤2 hours for urgent jobs** and median arrival **≤4 hours**.

5. Low literacy, low data and local languages require WhatsApp, USSD, voice and people

The reach constraints

Mobile phones reach nearly everyone, but the internet does not:

- **32.1 million cellular connections (104% of population) but only 14.3 million internet users (46.3%)**, with 16.6 million people offline, as of October 2026 (DataReportal 2027).
- Personal phone ownership is **59.4% of adults** (urban 70%, rural 45%) (INS ECAM5).
- Data costs **$1.63/GB** (Les Afriques).
- Adult literacy is about 78%, with rural and female deficits.

Conflict resolved. DataReportal's late-2025 edition gave 12.6 million users (41.9%) (DataReportal 2026). The series is revised between editions, so treat the levels as approximate: **roughly 42–46% online**.

Language follows geography. Douala is French-dominant, with Duala as its lingua franca and Camfranglais among youth. About half of Cameroonians speak Pidgin in some form (Wikipedia: Cameroonian Pidgin).

The HCI evidence is blunt. "Textual interfaces are unusable by first-time low-literacy users". That includes SMS and USSD menus. A **live operator was up to about 10x more accurate** than text interfaces (Microsoft Research, Medhi Thies et al.).

Recommended channel mix

Channel	Artisans	Customers	Cameroon provider route	Indicative cost
Field agents ("ambassadeurs de quartier")	Core onboarding, verification, photo/ID capture	Assisted booking, dispute calls	Recruited from associations, churches, mobile-money kiosks	Stipend + per-activation bonus (not sourced)
WhatsApp bot (Flows, voice notes)	Portfolio photos, voice job reports (smartphone owners)	**Core**: request, receive shortlist, rate by voice note	Meta Cloud API via pass-through BSP or AWS End User Messaging Social	Utility $0.004, marketing $0.0225 per template; user-initiated free (Meta; Whautomate)

Channel	Artisans	Customers	Cameroon provider route	Indicative cost
USSD	**Core**: register, availability toggle, accept leads, confirm job, check own score	Verify artisan by numeric ID, quick rating	Local VAS aggregator (LMT, Nexah) or #237# switch — **not Africa's Talking** (no Cameroon coverage) (AT)	10 XAF/session, 120 s max (Orange 2020) (ART catalogue); vendor quote 250k XAF/month/operator + 15 XAF/session (HelloDuty)
SMS	OTP, job alerts, completion codes	Job confirmation code, review request	Local aggregator (cheapest) plus AWS End User Messaging as second path	3.5–8 XAF wholesale (Orange 2020); about $0.035 retail (D7)
IVR / missed-call call-back	Audio menus in French/English/Pidgin, recorded voice profile	Inclusive line for basic phones	Local contact centre (LMT: 400+ multilingual agents) (LMT)	Per-minute rates not retrieved
QR + printed 6-digit ID	On vest, badge, workshop sign, sticker on installed fixture	Scan, or dial/SMS the ID	Local printing	One-off, not sourced
PWA (<1 MB first load)	Profile management	Public profile behind QR	Self-hosted	About 1–3 XAF per 2 MB page at MTN prices (MTN bundles)

The design rules that follow from the evidence:

- **One numeric artisan ID** works across QR, USSD, SMS and IVR.
- **Shallow menus:** two to three levels and about five options, to fit the 120-second USSD session.
- **Ratings by keypad.**
- **Recorded audio prompts, not text-to-speech.** AWS Transcribe, Polly and Lex, Google Chirp and Whisper cover French and English but **no Cameroonian Pidgin or local language** (AWS Transcribe; Google STT). Pidgin and local-language voice notes are stored as audio and tagged by human moderators, the CGNet Swara model (CGNet Swara).
- **Skip KaiOS.** Its web share in Cameroon is 0%, against 82.8% for Android (StatCounter).
- **Lean on cheap WhatsApp bundles.** MTN sells WhatsApp-only data from **25 XAF for 25 MB**, which makes voice-note and photo workflows affordable for artisans who own a smartphone (MTN).

Six channel facts need resolving:

- **Two-way SMS.** AWS says Cameroon supports short codes and two-way SMS. Twilio says two-way is not supported (AWS; Twilio). **Route inbound "reply 1/2" ratings through a local aggregator's short code.**
- **Sender ID.** Register it about 3–4 weeks before launch.
- **Orange's self-serve SMS API** is discontinued from 1 December 2026 (Orange Developer).
- **USSD access has been politically contested.** MTN suspended Express Union's code in 2017 and the minister intervened (Camer.be). Get written commitments and keep the #237# national switch as a fallback (ITWeb).
- **WhatsApp pricing changed on 1 October 2026.** Service replies above 1,000 a month per number are now charged, which favours user-initiated conversations.

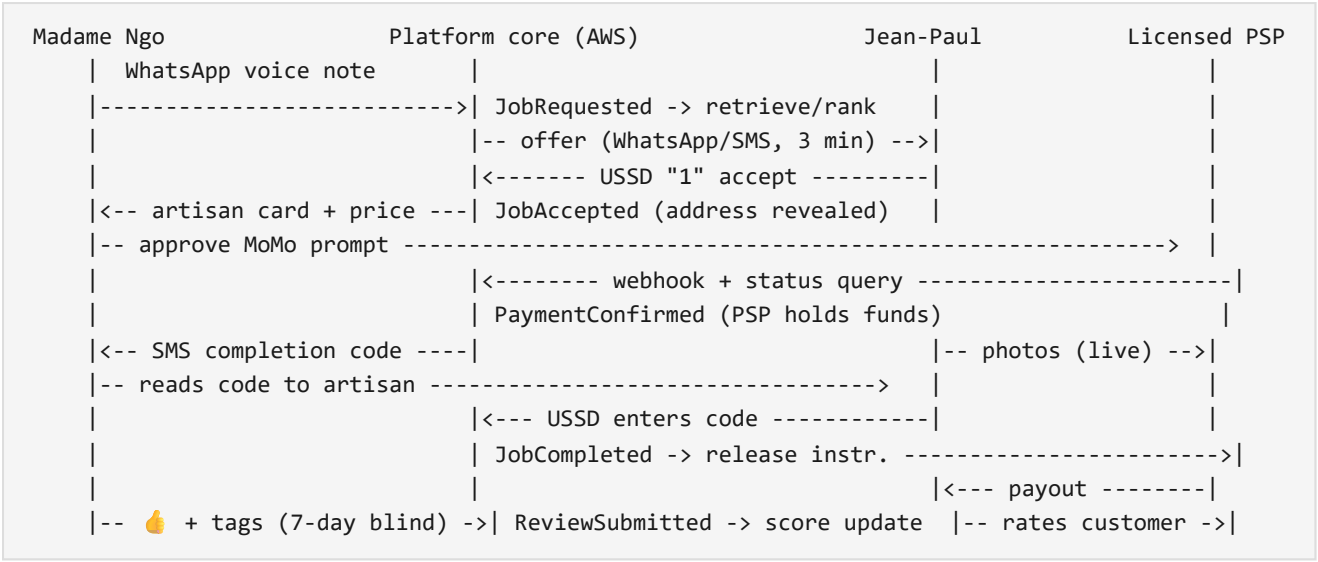
- **No reliable WhatsApp penetration figure** for Cameroon was found.

6. One end-to-end demo journey, from verification to recorded reputation

The demo should show **one real journey with evidence**, framed inside the "one trade, one quarter" story. Judges and partners reward a complete loop over breadth (Paul Graham; a16z glossary). Below, each step names the channel, the system action and the domain event the backend records.

Step	What happens (persona: Jean-Paul, electrician; Madame Ngo, Bonamoussadi)	Channel	Backend action and event
1. Register	Agent onboards Jean-Paul at a "Journée de vérification" at the association hall; phone OTP; mobile-money name match; live selfie with liveness; CNI or récépissé photo; trade and quarter	Agent PWA + SMS OTP	Cognito OTP; Rekognition liveness + face dedupe; ID image deleted after check; `ArtisanRegistered`, `IdentityVerified(level=0→1)`
2. Vouch	Association head and chef de quartier vouch (two circles) in person; workshop geotagged by agent	Agent PWA / USSD for vouchers	`VouchGiven(voucher, vouchee, circle, stake)`; vouch budget decremented
3. Skill test	45-minute standard task (wire and earth a circuit with breaker), scored by two masters on a checklist; video stored privately	Agent PWA	`SkillCertified(tier=3, assessors, scores)`; inter-rater check
4. Badge	Vest, laminated badge with QR + 6-digit ID printed on site	Print	Public profile URL; numeric ID live on USSD/SMS
5. Request	Madame Ngo sends a French voice note on WhatsApp: "Ma prise fait des étincelles" ("my socket is sparking"); shares location pin; agent-operator or rules classify "urgent electrical" and show price band	WhatsApp	`JobRequested(category, h3_r8, urgency)`; price band lookup

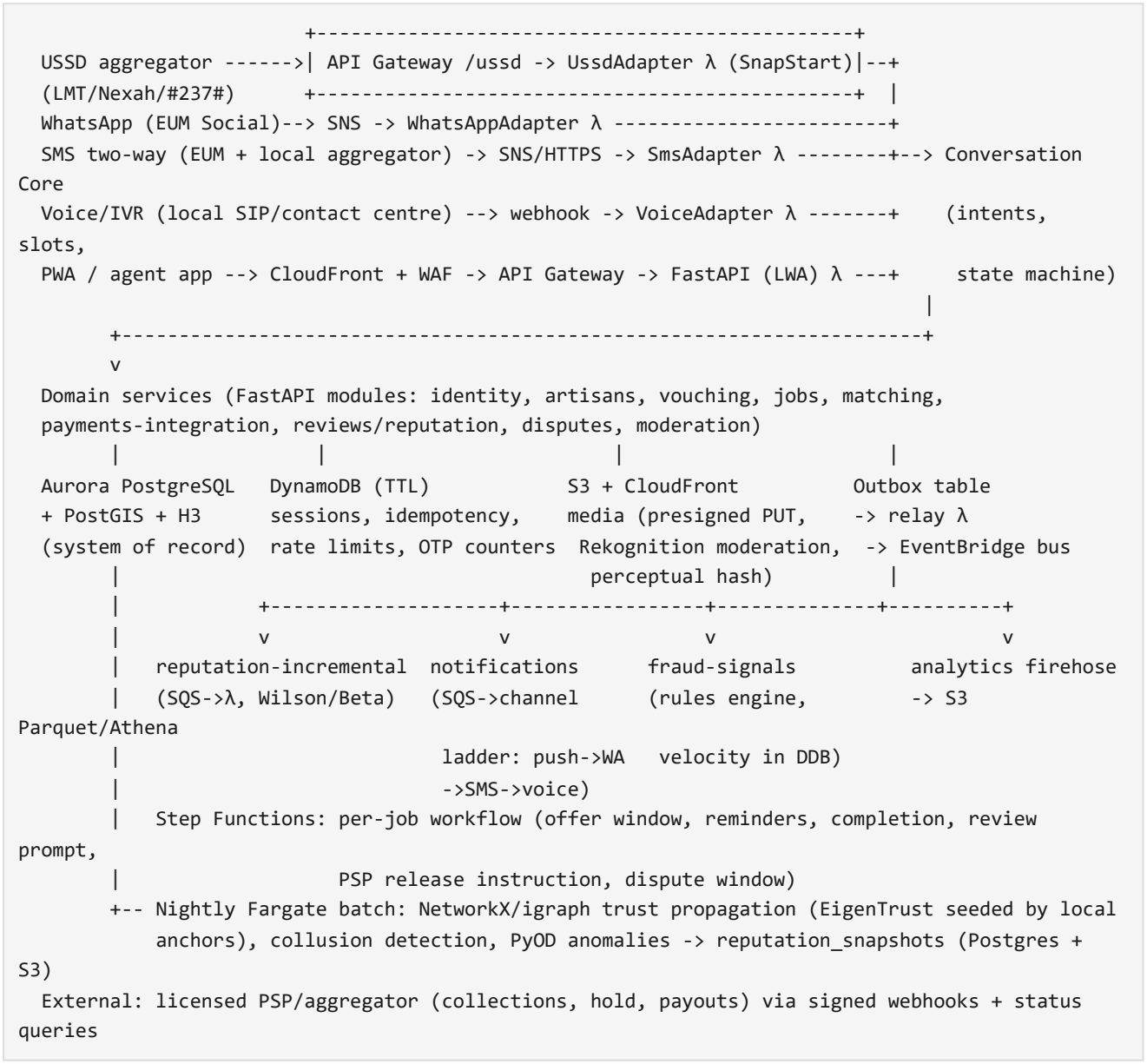
Step	What happens (persona: Jean-Paul, electrician; Madame Ngo, Bonamoussadi)	Channel	Backend action and event
6. Match	Top 3 verified electricians within ring k≤2 get a 3-minute offer by WhatsApp/SMS; Jean-Paul accepts on USSD ("1 = accept"); customer sees his card ("34 jobs checked ✓, 92% 👍, 5 people from your quarter hired him")	WhatsApp, USSD, SMS	Step Functions job workflow; `JobAccepted`; exact address revealed only now; numbers masked
7. Pay into protection	Customer pays labour + 10% protection fee via MoMo/Orange Money collection request through a licensed aggregator; funds held in **PSP's** ring-fenced account	MoMo/OM prompt	`PaymentConfirmed(psp_ref)` only after server-to-server status check
8. Do the job	Arrival selfie; live-camera "before" photo; repair; "after" photo; QR sticker placed on the breaker panel	WhatsApp/PWA camera	Photos hashed (perceptual hash), server-timestamped, coarse H3 r9; `JobStarted(geo, photo)`
9. Complete	Customer reads 6-digit completion code (sent by SMS at job start) to Jean-Paul, who enters it via USSD	SMS + USSD	`JobCompleted(completion_code_verified=true)`; release instruction sent to PSP; payout to Jean-Paul's wallet
10. Rate and record	Both sides rate within 7 days (customer: 👍 + "on time", "clean"; artisan: "paid as agreed"); simultaneous reveal; Jean-Paul's verified-jobs count and conservative score update; job appears on public profile as a "verified job"	WhatsApp / USSD keypad	`ReviewSubmitted`; incremental reputation update; nightly graph audit
11. Dispute path (one slide)	If the socket fails within 30 days: free return; else platform pays another artisan up to the guarantee cap; escalation platform → association → chief (if both agree)	Any	`DisputeOpened/Resolved`; voucher stake affected only if upheld



A few rules keep the demo honest. Use seeded but clearly labelled data, and disclose any Wizard-of-Oz operator. Faking activity carries "serious reputational risk" (NFX). Bring physical artefacts (vest, badge, sticker), show the SMS/USSD fallback, and keep a pre-recorded video in case of network failure.

7. Build it as a serverless Python modular monolith on AWS, cell-ready from day one

Architecture



The stack choices, with the evidence behind them:

- **API layer.** FastAPI, Pydantic v2, SQLAlchemy 2.x with GeoAlchemy2, and Alembic, deployed with the **AWS Lambda Web Adapter** so that "the same image runs on EC2, Fargate and locally" (AWS LWA).
- **Cold starts.** Lambda SnapStart for Python 3.12+ cuts init latency to as low as sub-second. Connections opened during init must be re-established after restore (Lambda SnapStart).
- **Database.** Aurora Serverless v2 can scale to zero in dev (Aurora auto-pause). Keep a minimum of 0.5–1 ACU in production to avoid the roughly 15-second resume.
- **Workers.** SQS-triggered Lambda workers with Powertools (idempotency, batch processing, structured logging) replace Celery (Powertools idempotency).

- **Build and deploy.** Package with uv lockfiles, define infrastructure in AWS CDK (Python), and deploy through GitHub Actions with OIDC rather than static keys (GitHub OIDC).

Several AWS services are closed, retired or missing in Cameroon:

- **Amazon Fraud Detector** closed to new customers on 7 November 2025 (AWS).
- **Amazon Pinpoint** reached end of support on 30 October 2026. Its messaging APIs continue as End User Messaging (AWS Pinpoint).
- **QLDB** was reported end-of-life (unverified). Do not build a "trust ledger" on it; use an append-only, hash-chained Postgres event table plus S3 Object Lock exports.
- **USSD** has no AWS service, so the aggregator is the one unavoidable non-AWS dependency.
- **Amazon Connect** is not offered in Paris or Ireland, and its outbound campaigns from Cape Town reach only South African numbers (Amazon Connect regions). Voice therefore goes through a local SIP or contact-centre provider.

Data and geography

Data	Store
Users, artisans, skills taxonomy, jobs, quotes, reviews, vouches, disputes, payment references, landmarks, price observations, consent receipts, outbox	Aurora PostgreSQL + PostGIS (`geography(Point)` + GiST; H3 `bigint` columns, B-tree) (Aurora PostGIS)
USSD sessions, WhatsApp cursors, OTP counters, idempotency keys, rate limits	DynamoDB on-demand with TTL
Proof media, portfolio	S3 (private, signed CloudFront URLs for proof; public cache for portfolio)
Events, analytics	S3 Parquet + Athena via outbox → EventBridge → Firehose
Graph analytics	NetworkX/igraph on Fargate nightly; Postgres recursive CTEs (2–3 hops) online; Neptune Analytics only if the graph outgrows memory

Cameroon has no street addresses, so addresses are modelled as **landmark + free-text directions + GPS point + accuracy**:

- USSD users pick city → quartier → landmark.
- WhatsApp users share a location pin.
- Landmarks are seeded from OpenStreetMap points of interest, and users can add local aliases.

Commercial maps are weak here. Amazon Location's Open Data provider cannot geocode or route, and GrabMaps covers only Southeast Asia (Amazon Location providers). OSM mapping of Douala's outskirts is incomplete: a 2026 UN Maps mapping pilot covered Yaoundé, and Douala is still only planned (UN Maps). Self-host OSRM or Valhalla if travel times are needed.

Reliability, offline tolerance and channel engineering

USSD is synchronous and needs replies within about 1–2 seconds:

- Keep that Lambda warm and free of database joins.
- Store session state in DynamoDB.

- Commit side effects asynchronously and confirm them by SMS.
- Persist drafts for 24 hours, because phones die during power cuts.

Webhooks will be retried: Meta retries for 36 hours (Meta Webhooks). Make them idempotent, keyed on channel + provider message ID, and use SQS FIFO grouped by MSISDN to keep each user's messages in order.

Fallback. AWS warns that SMS in each country runs through a single carrier partner (AWS SMS). Run two SMS paths and a fallback ladder: push → WhatsApp template → SMS → voice call.

Shutdowns. Make every critical step completable over SMS or USSD, so that a regional WhatsApp block does not break the trust loop.

Region choice

Conflict resolved. The system-design note (06) recommends eu-west-3 (Paris) on latency grounds. Cameroon's main international cables (SAT-3, WACS) land in Europe, and Camtel's Lagos link is reported at 92% utilisation (Telecom Review Africa). The security note (07) suggests af-south-1 (Cape Town), which keeps data on the continent under South Africa's POPIA and an operational regulator.

The legal analysis cuts both ways. **Both are cross-border transfers** requiring prior authorisation under Law 2024/017 Art. 32. The draft decree's adequacy test (a specific data-protection law plus an operational authority) is met by France and South Africa alike (MINPOSTEL draft decree).

Recommendation: run a one-week RTT probe from MTN, Orange and Camtel SIMs in Douala to both regions. Default to **Paris primary, Ireland for DR**, unless counsel advises that the APDP is likely to favour African hosting. Build cell-awareness into IDs, events and IaC from the start, so that moving or adding an af-south-1 cell is an infrastructure change rather than a rewrite. Region latency matters less than it seems, because most interactions are USSD, SMS or WhatsApp, whose latency is dominated by the network operators and Meta.

Scaling path and cost

Stage	Users	Shape	Infra cost (rough, excl. messaging)
MVP	≤10k	One cell (prod-cm) in eu-west-3; Lambda + Aurora Serverless v2 Multi-AZ + DynamoDB + EventBridge/SQS/Step Functions + S3/CloudFront	~$100–250/month at 1k MAU
Growth	10k–300k	RDS Proxy, read replicas, ElastiCache, data lake, nightly Fargate batch, learning-to-rank; new cities = new `market` rows	~$1.5k–3k/month at 100k MAU
Scale	300k–1M+	Main API to ECS Fargate/Graviton (same image); partition large tables	~$10k–22k/month at 1M MAU
Multi-country	1M+	Cell per country/regulatory bloc; thin global control plane; route by E.164 prefix (+237, +241, +242…)	Per cell

These cost figures are order-of-magnitude estimates from note 06, built on us-east-1 list prices. **Messaging, not infrastructure, will dominate**: per-user SMS, WhatsApp template and USSD fees can run around 10 times the per-user infrastructure cost. New AWS accounts get up to $200 in credits, and Activate Portfolio offers up to $100k–200k through an accelerator (AWS free plan; AWS Activate). The cell-

based design follows AWS's own guidance: no shared databases across cells, deployed in waves (AWS cell-based architecture).

8. Local trust is borrowed from associations, chiefs, churches, njangis and hardware stores

The launch evidence converges on one rule: **win the hard side first inside a tiny atomic network**. In Lenny Rachitsky's study, 14 of 17 marketplaces concentrated early effort on supply (Lenny's Newsletter). Thumbtack's founder says over 90% of local-services GMV still moves by word of mouth (NFX: Thumbtack). Here the hard side is not artisans in general, who are plentiful, but **good, responsive, honest** artisans who accept verification and keep jobs on-platform.

Trust in Cameroonian neighbourhoods already runs through institutions:

- **Chiefs** are legally "auxiliaires de l'administration" with a tradition of mediation (Wikipedia FR: Chefferie).
- **Churches** reach about two-thirds of the population (Wikipedia: Religion in Cameroon).
- **Njangi groups** rest on members choosing each other on trust (Wikipedia: ROSCA).
- **Radio** is a mass channel, with many private stations in Douala (Wikipedia: Media of Cameroon).

M-Pesa's lesson is to recruit existing trusted retail nodes as agents, and that agents' trust in the system comes before customers' trust (Wikipedia: M-Pesa).

Partner	Role	Offer in return
Electricians' association / trade group	Supply gatekeeper, co-verifier, assessors, co-branded badge	Data, visibility, per-test fee
Chef de quartier	Endorse at quarter meeting; first-instance mediator; residence attestation	Recognition; free priority service for community buildings
Parishes / assemblies	Demand trust: announcements, fellowship referral codes	Group referral rewards
Njangi and women's groups	Referral; "njangi-backed" vouch	Reward paid into the njangi pot
Quincailleries (hardware stores)	Daily artisan footfall; onboarding desk; "buy the part, get a verified installer"	Referral fee, footfall
Mobile-money kiosks	Payment explanations, onboarding points	Per-activated-user fee
Landlords, estate agents, property managers	First recurring B2B demand	Priority dispatch, monthly invoicing, photo proof
Douala FM / church radio	"Ask the electrician" call-ins in French/Pidgin	Content

Physical trust artefacts make the platform visible: branded vests, a laminated badge with photo, QR and numeric ID, and **QR stickers on installed fixtures** ("installed by X, verified by [platform], scan to rate or rebook"). The sticker doubles as an anti-leakage and rebooking device. Publish electrical-safety outcomes, such as earthing and breakers installed, as a public-good metric. The July 2026 Bonamoussadi collapse makes workmanship a live local concern.

The Douala-first sequence matches Yango, which started in Douala before Yaoundé (Wikipedia: Yango), and Gozem, whose Cameroon office is in Douala (Gozem). Gozem's zero-deposit vehicle financing, repaid

from earnings, is the most transferable Cameroon-tested idea. The artisan version is **tool or kit credit repaid as a small deduction from on-platform payouts**, which also discourages leakage.

9. Anti-fraud means anchoring reputation to proven, paid, co-located jobs by unique people

The threat is mostly self-boosting

Fake reviews pay, which is why people buy them:

- Facebook-group review schemes on Amazon raised ratings about 5% and doubled review counts. After sellers stopped buying, ratings fell from 4.4 to 4.1. Manipulation is used mainly by low-quality products (He, Hollenbeck & Proserpio).
- Weak-reputation independents with few reviews in competitive markets manipulate most (Luca & Zervas). That describes almost every Cameroonian artisan.
- At Tripadvisor, "review boosting" by owners or staff was **54% of fraud** (Tripadvisor).
- Trustpilot removed 4.5 million fakes (7.4% of submissions), most of them 5-star (Trustpilot).

Verified purchase alone is not enough. Amazon reviewers bought the product and were refunded off-platform. An artisan can do the same: pay friends to book through the platform, refund them by MoMo, and collect "verified" reviews.

Cameroon adds weak phone identity and a professionalised scam economy:

- One national ID card was found linked to **up to 200 SIM cards** in 2019 (Africanews), and 891,082 non-compliant SIMs were cut in 2022 (ITWeb).
- Prisons are named as cybercrime hubs (Digital Business Africa).
- Cybercrime cost XAF 12.2 billion in 2021 (EcoMatin), and only **6 in 100 victims file a complaint** (EcoMatin).
- MoMo fraudsters trigger withdrawal requests that careless users approve with their PIN (Camer.be).

With so few complaints filed, **the platform's own one-tap reporting is its main fraud sensor**.

Threat model and controls

Attacker	Method	Primary controls
Artisan	Sockpuppet customer accounts	Device graph + Play Integrity verdicts; one review per verified job; duplicate-face check on artisans; device-recall blocklist (Play Integrity)
Artisan	Friends-and-family reviews	Platform-routed payment weighted higher; reviewer-diversity metric; diminishing weight for repeat pairs; **down-weight, don't delete**
Artisan	Refunded fake jobs	Wallet-flow checks (artisan→reviewer transfers); GPS co-presence + minimum time on site; live photos
Artisan rings	Reciprocal reviews, vouching cartels	Cycle detection; dense-subgraph (FRAUDAR-style) scoring (FRAUDAR); vouch-budget limits
Competitor	1-star bombing	No job, no review; spike detection + review freeze with banner (Google pattern) (Search Engine Roundtable)

Attacker	Method	Primary controls
Customer	"Discount or 1 star" extortion; non-payment	Simultaneous reveal; extortion flag with in-app chat evidence; no review without payment status; rater-reliability weighting
Scammer	Fake artisan taking deposits	Tier-1 KYC (CNI/récépissé + liveness + face dedupe); deposits only via PSP hold; pinned "never pay a deposit outside the app" warning
Scammer	Account takeover via SIM swap / insider SIM	Selfie re-verification on number/device change; payout-wallet change cool-off and step-up; alerts to old number
Broker	Selling reviews or aged accounts on WhatsApp/Facebook	Monitor groups, test-buy, report to Meta; account-age features
Collusion	Customer + artisan fake "verified" jobs	Graph + payment anomaly models; random field-agent call-backs

Staged detection on AWS and Python

Detection should grow with the data:

- **Stage 1 (launch, no labels).** A versioned rules and velocity engine in Python. Counters live in DynamoDB or Redis. Example rules: more than N accounts per device or per hashed CNI; review posted within minutes of job close; no handshake; empty Play Integrity verdict; payment followed by a reverse transfer. A nightly NetworkX job adds connected components on shared identifiers, reciprocity cycles, reviewer-set overlap, SybilRank-style propagation from ID- and payment-verified accounts (SybilRank), and PyOD/IsolationForest anomaly scores (PyOD).
- **Stage 2 (hundreds of labelled cases).** AutoGluon or LightGBM on SageMaker, which AWS names as Fraud Detector's replacement. Graph neural networks (Neptune ML, PyG) come only then (Neptune ML).

Identity checks use Rekognition Face Liveness at about **$0.015 per check** (Rekognition pricing), or Smile ID, which supports six Cameroonian document types and duplicate-face detection but not government database lookup (Smile ID). Accept récépissés and old-format cards: the biometric CNI launched only in February 2025 and had backlogs (Biometric Update). Use the hashed 17-digit CNI unique identifier as a uniqueness key (Osiris/ART). Keep customer friction low: phone OTP plus a device check, with ID requested only when risk signals fire.

Governance follows a graduated penalty ladder:

1. Silent down-weighting.
2. Education.
3. Warning.
4. Public "unusual activity" badge (Yelp-style).
5. Score frozen (Trustpilot-style).
6. Suspension.
7. Ban, with face and device blocklisting.

Appeals go to a different human within 14 days, with a target of 72 hours. Publish a "how reviews are verified" page and a periodic trust report, following the UK CMA's "reasonable and proportionate steps" standard (CMA208) and the EU's disclosure rule (Mondaq).

Conflict resolved. The anti-fraud note (04) proposes features such as "same surname or locality" and contact-list uploads to detect friends-and-family collusion. Surnames and origins can act as proxies for ethnic or regional origin, which Law 2024/017 Art. 48 prohibits processing. **Drop surname-based features and contact uploads.** Rely on device, payment-flow, timing and graph-structure signals instead.

10. Security, data protection and legal compliance set hard design limits

Law 2024/017 is strict and its regulator is missing

Cameroon's data-protection law took effect on 23 December 2024, and its **18-month grace period ended on 23 June 2026** (Law 2024/017 official text; Blacktree). Its main obligations:

- **Consent is the default legal basis** (Art. 9). There is no contract or legitimate-interest ground.
- **Processing needs prior authorisation** (Art. 49).
- **Every transfer abroad needs prior APDP authorisation** (Art. 32). Fines are 10–50M FCFA (Art. 60), and criminal exposure is 3–10 years (Art. 69).
- **Breaches must be notified "sans délai"** (Art. 22).
- **Data subjects have a right to human intervention** in automated decisions (Art. 44).
- **Legal persons face criminal fines of up to XAF 1 billion** (Art. 71).

But the presidential decree creating the authority (APDP) had not been confirmed as adopted as of October/November 2026 (DLA Piper; Kukie.io). The authorisations are legally required but practically impossible to obtain.

Recommendation: keep a dated compliance file covering the processing register, DPIA, transfer impact assessment, consent receipts and a letter documenting good-faith filing attempts. Submit filings the day the APDP opens. Ask counsel about interim routes via MINPOSTEL (lawyer).

The profiling problem: Art. 65 versus reputation scoring

This is the report's most important legal conflict. Notes 03 and 06 design automated trust scores, ranking and fraud models. Art. 65 punishes "processing personal data for profiling purposes" with **3–10 years and XAF 1–20 million**, and the text has no explicit consent carve-out. One commentator wrongly claims the law does not address automated decisions; Art. 44 shows it does.

The design response has six parts:

- **Score service events, not people.** Use verified jobs, completion, disputes and vouches. Never use demographics or inferred traits.
- **Consent separately.** Collect a distinct, recorded consent for "publication of my job record and trust badge".
- **Explain the score.** Store component scores so any channel can say *why*: "12 verified jobs, 3 vouches, 0 disputes".
- **Keep humans in the loop.** No automated suspension. Human review and contestation under Art. 44.
- **Keep the badge simple.** Prefer simple, auditable rules for badge tiers over opaque ML in the public score.

- **Get a written legal opinion first.** Launch automated scoring only after counsel has signed off (lawyer). Until then, the concierge pilot can show verified-job counts and human-assigned tiers, which are much harder to characterise as "profiling".

Art. 48 adds a second limit. It prohibits processing data on **"linguistic or regional origin"** and ethnicity. The reputation note's monthly bias audit "by gender, age, region and language group" cannot be run on profile attributes. Preferred language must stay a UI setting, never a matching or ranking feature. Any fairness audit should use separately consented, aggregated survey data designed with counsel.

Biometrics are a third. The law's text targets "biometrics in health matters", but commentators treat biometrics broadly, and Art. 66 criminalises storing sensitive data without express consent. So: separate explicit consent, a licensed KYC vendor or Rekognition, storage of templates (face vectors) rather than images, and deletion of ID and selfie images after verification.

Payments: the escrow conflict

This is the second key conflict. The business-model note (08) proposes that the customer pays "into the platform's collections account", with funds held until confirmation or a 48–72-hour auto-release. Regulation 04/18/CEMAC/UMAC/COBAC rules this out:

- Payment services, including money remittance, executing transfers and issuing e-money, are reserved to licensed credit, microfinance and payment institutions (Arts. 3 and 5).
- Client funds must be ring-fenced at a CEMAC bank by the next day (Arts. 52–54) (Regulation 04/18).
- The BEAC's own 2024 report describes fintechs as "generally technical or commercial partners of licensed establishments, acting as payment aggregators" (BEAC 2024 report).

Resolution: the platform never holds customer money. A licensed PSP or aggregator (for example pawaPay, an MNO merchant collections account, or a bank-backed aggregator) collects from the customer into **its** ring-fenced account. It holds the funds per the platform's job state and disburses to the artisan when it receives a signed release instruction. The platform's fee is either deducted by the PSP or invoiced separately. The platform acts as a merchant or "partenaire technique", not a payment provider.

Three cautions apply. MTN has not publicly documented a native marketplace split-payment product. A developer's May 2026 question about 90/10 splits went unanswered (MTN MoMo community). So **confirm hold-and-release capability in the PSP contract** (lawyer: whether any platform-held balance or wallet triggers licensing or distributor status). Cameroon remains on the **FATF grey list** as of October 2026, which shapes partners' KYC posture (FATF). And never trust a payment callback without a server-to-server status query.

Other law that applies

Area	Rule	Design response
E-commerce law 2010/021	Transaction-data retention (Art. 32); opt-in for automated direct marketing (Art. 7)	Separate transaction records from profile data; marketing opt-in only (also WhatsApp policy)
Cybersecurity law 2010/012; ANTIC	National ICT security; applicability of audits to private marketplaces unverified	Lawyer: ANTIC obligations, incident reporting
Consumer law 2011/012	Information duties, unfair terms (texts not retrieved)	Terms state the platform is an intermediary; artisan warranties; no total liability exclusion

Area	Rule	Design response
Labour Code (Law 92/007)	Misclassification risk if platform controls workers	No mandatory schedules, exclusivity or uniforms as conditions; ratings framed as marketplace integrity; lawyer
Tax	VAT 19.25%; CIT 33% (27.5% reduced); **minimum tax 2.2–5.5% of turnover** (PwC; PwC CIT)	Agency/commission structure so turnover = platform fees, not GMV; accountant
WhatsApp policy	Opt-in; no collection of ID numbers or bank numbers in chat; human escalation (WhatsApp policy)	KYC uploads via secure link or Flow, never chat text
Insurance	CIMA-licensed insurers for liability and personal-accident cover	Price a guarantee reserve and embedded accident cover with a broker

Security blueprint

The account and identity structure:

- **Landing zone.** Control Tower with a management account (billing only), Security Tooling (delegated admin for GuardDuty, Security Hub, Inspector and Macie), Log Archive (organisation CloudTrail with S3 Object Lock), and prod, staging and dev accounts (AWS SRA).
- **Guardrails.** SCPs deny regions outside primary and DR, which also produces evidence for the transfer file, and deny disabling logging.
- **Staff access.** IAM Identity Center with passkeys.

Authentication rests on Cognito passwordless SMS OTP (Cognito). NIST classes SMS OTP as **"restricted"** and vulnerable to SIM swap (NIST SP 800-63B-4), so harden it:

- 10-minute expiry and at most 5 tries per code;
- WAF rate limits per IP and per phone number, because Cognito threat protection "doesn't apply rate limits" (Cognito threat protection);
- SMS country allow-listing to +237 to block SMS-pumping fraud;
- step-up (passkey, or OTP plus PIN) for payout-wallet changes, the main SIM-swap target.

Application security follows **OWASP API1 (Broken Object Level Authorization)**, the dominant marketplace risk (OWASP API Top 10):

- one policy layer for object checks, with UUIDv7 identifiers;
- artisans never see a customer's full phone number or address before acceptance;
- pip-audit and Bandit with hard fails in CI (pip-audit);
- verification of Meta's X-Hub-Signature-256 and PSP signatures over the raw request body;
- no personal data and no payout changes over USSD.

Privacy by design for low-literacy users means:

- **Voice consent.** Layered 30–60-second audio consent in French, English and Pidgin, with recorded keypress receipts.
- **Minors.** Parental consent for under-18 apprentices, or exclude them.
- **Location.** Grid-snapped artisan locations, with the customer's exact address revealed only after acceptance and expired after completion.

- **Contact.** Masked numbers.
- **Safety.** SOS and "share visit" features. Technology-facilitated violence affects an estimated 28% of women in five Sub-Saharan countries (UN Women).

Incident response follows NIST SP 800-61r3 and is wired to Art. 22. Target notification within 72 hours as a defensible practice, with pre-drafted French and English notices plus SMS and voice versions (NIST SP 800-61r3). Start with a security.txt disclosure policy (RFC 9116). Move to a bug bounty only once triage is mature, and to SOC 2 or ISO 27001 when PSP or enterprise partners require it.

11. A Garantie-first model monetises customer risk, not artisan income

Why lead fees and subscriptions are out

Lead fees (Thumbtack, Bark, MyBuilder) and subscriptions (Checkatrade, from £30 a month on 12-month terms) charge tradespeople before they earn, which breaks the requirement (Housecall Pro; Whito). Commission models fail when the platform does not control payment. Thumbtack's first commission model died because customers contacted providers directly (Sharetribe). Customers *will* pay explicitly for protection, as TaskRabbit's Trust & Support fee shows (TaskRabbit).

Cheap peer-to-peer mobile money changes what customers will pay for. Both operators now offer zero-fee transfers plus taxes, and withdrawals are 1% in-app (EcoMatin; Digital Business Africa). Customers can already pay artisans cheaply, so **the platform must sell protection, not convenience**.

The African record points the same way:

- Lynk's full-stack model, with in-person vetting, an academy and a 10% fee, wound down and was sold to Eden Life in 2022 (TechCabal). Eden paused its consumer business in February 2026 (TechCabal).
- Asset-light Kandua, with free tools for pros and B2B channels, reached **40,000+ vetted firms** and was bought by insurer Santam (Santam).

Recommended revenue stack

Stream	Mechanism	Timing
Customer protection fee	8–10% of labour, min XAF 500; funds PSP-held hold-and-release, rework/refund guarantee (cap e.g. XAF 50,000 in pilot)	Month 0
Success commission	0% at launch; 5–8% deducted **only from released payouts** after the artisan's 3rd paid job; lower on repeat clients	Month 6+
B2B contracts	Landlords, property managers, SMEs, hotels: retainers + per-job margin, monthly invoicing, photo proof	Month 0 pilots
Diaspora milestone projects	5–10% fee on XAF 0.5–5M renovation/build tickets; inspector photos per milestone	Month 6+
Ancillary	Materials affiliate (2–5%), MFI loan and micro-insurance referrals, sponsor-funded certification cohorts, optional "Pro" visibility paid only from the earnings wallet	Year 2

Unit economics

Note 08's illustrative model for a household job is below. All inputs are assumptions except the payment and tax rates. Note the added column for **year 1**, when the success commission is 0%.

Line (per job, XAF)	Steady state	Year-1 launch (0% commission)
Labour ticket	15,000	15,000
Customer protection fee (10%)	1,500	1,500
Artisan success commission (5%)	750	0
Gross take	2,250	1,500
Net of 19.25% VAT (if fees VAT-inclusive)	~1,887	~1,258
Collection cost (~3% of 16,500)	−495	−495
Payout cost (~1%)	−142	−150
Guarantee reserve (2% of GMV)	−300	−300
Support and verification	−300	−300
Contribution	**~650 (≈€1)**	**~13 (break-even)**

Payment costs come from pawaPay's published rates: collections 1.75% (MTN) and disbursements 1.3% (MTN) (pawaPay). On top come the 0.2% transfer tax and XAF 4 levy, which raised XAF 42.7 billion in 2025 (EcoMatin). Messaging adds an estimated XAF 20–60 per job: two or three WhatsApp utility templates at about $0.004, one or two SMS codes, and a few USSD sessions at 10–15 XAF. Model this explicitly.

The year-1 column shows **household jobs alone are break-even at best during the free-to-artisan launch**. B2B and diaspora streams must carry fixed costs from the start, and the commission switch-on in month 6 is what makes household volume contribute.

Lifetime value is low. A customer doing three jobs a year for two years is worth about XAF 3,900 in contribution, which caps blended acquisition cost at about **XAF 1,300–2,000 (€2–3)**. Demand must therefore come from artisans bringing their existing clients (escrow and reputation benefit them), WhatsApp referral credits, partner channels and field agents paid per verified first booking. Only targeted Facebook ads fit the budget.

Urban Company shows the ceiling and the trap. A heavily managed, standardised model reaches a 28–35% take, but its low-ticket InstaHelp line lost ₹346 per order (Inc42).

Market size and projection (assumption-based)

Douala's metro population is about 4.35 million (Wikipedia: Douala). Assuming about 870k households spend XAF 30,000 a year on electrical, plumbing and appliance-repair labour gives a **TAM of about XAF 26 billion**. A digitally reachable share of about 35% gives a **SAM of about XAF 9 billion**. A three-year **SOM of 3–5%** gives **XAF 270–450 million GMV** from household jobs. At a 15% take, that is only about XAF 40–70 million in revenue, so B2B and diaspora carry the plan.

Remittances anchor the diaspora case. **Conflict resolved:** note 08 cites $787 million at 1.6% of GDP. Note 01 shows that was the 2023 figure, and 2024 fell to about **$690 million (1.3% of GDP)** (Statistics of the World). The diaspora sizing should use about $690 million.

Driver (assumptions)	Year 1	Year 2	Year 3
Verified active artisans	250	900	2,000
Household jobs/month (end of year)	600	2,500	6,000
Household GMV (XAF)	~60m	~300m	~900m
B2B accounts / GMV	3 / 10m	12 / 60m	30 / 150m
Diaspora projects / GMV	10 / 15m	50 / 75m	120 / 180m
Blended net revenue (XAF)	~15m	~70m	~200m
Team (FTE)	5	10	16

The first-24-month funding stack should be non-dilutive and catalytic: founder and angel money, TEF's $5,000 grants (TEF), AWS and Google credits, and a sponsor-funded certification contract with a donor or insurer. A $150–500k pre-seed follows once there is GMV traction. Cameroon is barely visible in African VC data: $3.8 billion was raised across Africa in 2025, concentrated in South Africa, Kenya, Egypt and Nigeria (Briter).

12. Launch as a 12-week concierge pilot, then expand ring by ring behind hard gates

The pilot

Run a **concierge MVP over WhatsApp**:

- **Scope.** One trade (electricians), one quarter cluster (Bonamoussadi/Makepe/Kotto), 12 weeks plus 2 weeks of setup.
- **Supply.** 25–40 artisans verified at one or two "Journées de vérification".
- **Demand.** A target of 150–300 job requests from 100–200 customers, including 3–5 landlord or agency accounts. Enough to estimate fill and repeat rates within about ±7–10 points.
- **Tooling.** A WhatsApp Business number, a sheet or Airtable as the matching engine, a web profile per artisan behind the QR, PSP-collected payments, and ratings by voice note or poll.
- **Automation.** Automate only after about 100 manual matches show which steps repeat. Thumbtack regretted automating late (NFX: Thumbtack).
- **Activation.** Founders route each artisan's first two or three jobs within 30 days. Activation to first job is the key supply milestone (Andrew Chen).
- **Budget.** The researcher's unsourced estimate is **XAF 3–8 million** for 2–3 field ambassadors, badges, vests, stickers, a verification day, a demo day, radio spots, the guarantee reserve and data. Treat it as an estimate.

Each verification day runs six stations:

1. ID and mobile-money name match.
2. Residence attestation from the chief or landlord.
3. Two references, from past customers or the artisan's patron.
4. Practical test.
5. Photo and profile.

6. Code-of-conduct signature: transparent pricing, guarantee rules, dispute process.

Metric	Go threshold (proposed, unsourced)
Fill rate	≥80%
Median time to match	≤2 h urgent, ≤24 h planned
Job completion (matched → completed and paid)	≥85%
Dispute rate	≤5%, resolved within 7 days
90-day customer repeat	≥25%
Artisan 60-day retention	≥70%
Proof-of-work coverage (photos + rating)	≥90%
Referral/partner share of new customers	≥40%
Contribution per job after guarantee	≥0
Escrow uptake and protection-fee acceptance	Measure; the single most important unknown

Expansion rings

Local network effects reward density before breadth (NFX expansion framework). Move to the next ring only after **eight consecutive weeks** at the go thresholds:

1. **Ring 0 (months 0–3):** electricians in the pilot cluster.
2. **Ring 1 (months 3–9):** add plumbing, then painting and tiling, in the same quarters; then adjacent quarters such as Logpom.
3. **Ring 2 (months 9–18):** Douala-wide in three to five trades, codified in a "Quarter Launch Kit" (partner templates, verification-day script, ambassador pay, radio scripts, KPI dashboard).
4. **Ring 3 (18–30 months):** Yaoundé, a strong customer base with a 64.6% salaried share and an existing competitor presence, then Bafoussam and Kribi. Buea/Limbe only if lockdown risk is acceptable. Run each new city through a city-launcher with equity or revenue share.
5. **Ring 4 (30+ months):** CEMAC markets (Libreville, Brazzaville/Pointe-Noire), which share the CFA franc, French, and BEAC/COBAC rules (Wikipedia: CEMAC). Then WAEMU francophone markets, where the currency and rules differ. Each new country gets its own AWS cell.

The cautionary cases point the same way. Homejoy's 31-city sprawl and $19 discount cleanings kept only 25% of customers after month one (Harvard D3). Jumia suspended Cameroon e-commerce in 2019 (Wikipedia: Jumia).

Leakage risk is real but smaller for one-off urgent repairs than for recurring cleaning. The best direct measurement found disintermediation in no more than 1 in 83 relationships (Astashkina et al.). But stronger trust signals *can* raise leakage (Gu & Zhu). The platform must therefore keep adding value after the match: guarantee, payment protection, a reputation record that grows only from on-platform jobs, tool credit and insurance.

13. Risks, open questions and field-validation gaps

#	Risk or open question	Why it matters	How to resolve
1	Does Art. 65 (profiling) apply to consented, service-quality trust scores?	3–10-year criminal exposure for core product	**Written legal opinion before automated scoring**; human-assigned tiers in pilot (lawyer)
2	APDP not operational; transfer (Art. 32) and processing (Art. 49) authorisations unobtainable	Hosting abroad is technically unauthorised	Compliance file, good-faith letter, interim route via MINPOSTEL (lawyer)
3	Does PSP-held hold-and-release, or any platform balance, trigger payment-institution or distributor status under 04/18?	Licensing exposure	COBAC/BEAC counsel; PSP contract (lawyer)
4	PSP capability for marketplace hold/split in Cameroon undocumented	Core Garantie mechanism	Vendor due diligence (pawaPay, MNO merchant, bank aggregator)
5	Customer willingness to pay a ~10% protection fee; escrow uptake	Entire revenue model	Pilot A/B on fee level (8% vs 10%)
6	Leakage on repeat local jobs	Erodes commission	Track leakage proxy; lower repeat-client rate; sticker rebooking
7	Will associations, chiefs and churches act as vouchers/assessors, and on what terms (fees, liability)?	Tier 1 and Tier 3 depend on them	Field interviews; MoUs before pilot
8	Elite capture or partisanship of vouchers	Excludes outsiders, displaced anglophones	Two-circle rule, appeals, alternative voucher circles
9	Platform liability when a verified electrician causes a fire	Reputational and legal	ToS intermediary role, artisan warranties, CIMA-insured guarantee (lawyer/broker)
10	Labour misclassification from quality controls	Labour Code / CNPS exposure	Contract design (lawyer)
11	Minimum tax on turnover if revenue booked as GMV	Could exceed margin	Agency structure (accountant)
12	Voice consent validity as "univoque" evidence	Consent is the only legal basis	Lawyer; layered audio + keypress receipts
13	Electrician supply count and density in Douala V; neighbourhood demand, incomes, repair spend	Pilot sizing	Field scan; compare against Yaoundé (Biyem-Assi, Bastos)
14	Real XAF price lists per job; SMIG amount (36,270 in 2014 vs ~41,875 reported)	Price bands, unit economics	30–50 interviews; WageIndicator/official text
15	USSD and IVR tariffs on MTN, Camtel, Nexttel; #237# for non-financial services	Channel costs	Aggregator quotes
16	WhatsApp penetration; smartphone vs feature-phone split among artisans	Channel mix	Pilot intake survey
17	RTT from Cameroonian networks to Paris vs Cape Town	Region choice	One-week probe
18	Accuracy of French/English ASR on Cameroonian accents; no Pidgin ASR	Voice reviews	Pilot test; human moderation
19	Generalising the *Nature* binary-rating fairness result to Cameroon	Rating design	A/B binary vs 5-point in pilot

#	Risk or open question	Why it matters	How to resolve
20	Anglophone crisis, shutdowns, election-period throttling	Expansion, availability	SMS/USSD completeness; region SLOs
21	Competition from QwickServices and others	Differentiation	Depth in one trade and one area
22	Whether MINEFOP would accept platform portfolios as VAE evidence	Long-term moat	Early data-format dialogue with MINEFOP

Several **numbers in the notes were explicitly unverified and are excluded** from this report's claims:

- the "85% electrical fires" figure;
- Kolonell's vendor claims about Dakar (20% commission, 35% WhatsApp leakage, escrow cutting disputes 45%);
- Sent.dm's implausibly low SMS prices;
- 2025 ANTIC complaint counts that could not be tied to an article.

Several numbers **conflict between sources**, and this report treats them as follows:

- **Internet users:** approximate ranges, since DataReportal revises its series between editions.
- **Mobile-money market shares:** "roughly even". ART counted Orange at 50.08% of mobile subscribers in 2024 (EcoMatin/ART), Orange held 52.6% of mobile-money value in 2025 (Financial Afrik), and a vendor estimate gives MTN 52%. **Support both wallets from day one.**
- **NCSCS cable landing point:** Kribi versus Limbe, unresolved.
- **AWS Activate Portfolio credits:** $100k versus $200k depending on the page; verify when applying.

Conclusion

The research changes what "building the platform" means. The software is the easy part: a FastAPI-on-Lambda modular monolith with PostGIS, H3 and an event outbox can be built and run for a few hundred dollars a month. The defensible asset is a **verified job ledger**. Every entry is a job proven by a customer's code, a PSP payment reference and live photos, and is vouched for by people whose own standing is at stake. That ledger serves customers' trust and also anchors credit, insurance, VAE certification and B2B contracts, the revenue lines that do not tax artisans.

The binding constraints are legal and social, not technical. The platform must route money through a licensed PSP rather than hold it. It must keep reputation scoring narrow, consented, explainable and contestable until a lawyer clears Art. 65. And it must borrow trust from associations, chiefs, churches and njangis without letting any of them become a gatekeeper.

The most consequential unknown is whether Douala customers will pay roughly 10% for a guarantee when peer-to-peer mobile money is now nearly free. That single pilot metric decides whether this is a business or a public good. The electrician-safety framing and the diaspora and B2B channels are the strongest hedges if it turns out to be the latter.

14. Consolidated bibliography (801 distinct sources)

This list merges every source cited across the nine research notes (01 problem context, 02 comparable platforms, 03 skill verification and reputation, 04 anti-fraud, 05 channels and low-literacy UX, 06 system

design on AWS/Python, 07 security and legal, 08 business model, 09 launch and go-to-market). It was de-duplicated by URL: protocol, "www." and trailing slashes were ignored when matching. Where the same document appears on two different hosts (for example a Fairwork report on fair.work and on fairwork.oii.ox.ac.uk), both URLs are kept. The tag after each entry shows which notes cite it. Inclusion here does not mean the source is reliable. Vendor blogs, search-result summaries and other low-confidence sources are flagged in the body of this report where they are used, and the notes flag them too.

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